Characteristic-locus Design for the AIRC and TGEN Models

Multivariable Control, Take-home Exam

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Task

Carry out the characteristic-locus design procedure on two standard models from Maciejowski, *Multivariable Feedback Design*:

- AIRC, a 5-state aircraft model with 3 inputs and 3 outputs (`Q1_final.m`);
- TGEN, a 6-state turbo-generator model with 2 inputs and 2 outputs, designed for a lower bandwidth (`Q2_final.m`).

Method

The compensator is built in three frequency bands:

1. **High frequency, K_h.** A real constant approximation of $G(j\omega_b)^{-1}$ by the ALIGN algorithm. From the SVD $G(j\omega_b) = U\Sigma V^*$, take $K_h = -\operatorname{Re}(UV^*)$, which reduces misalignment at the design frequency ($\omega_b = 3\,\text{rad/s}$ for AIRC, $0.5\,\text{rad/s}$ for TGEN).
2. **Medium frequency, K_m.** A lead compensator on each channel to improve phase margin.
3. **Low frequency, K_ℓ.** PI action on each channel for zero steady-state error.

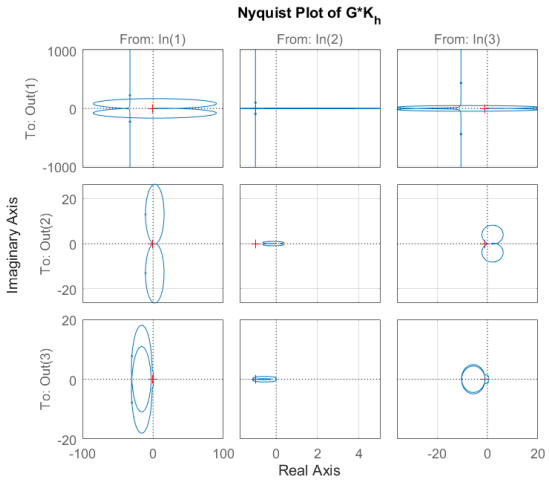
After each step I plotted the characteristic loci of GK, then the closed-loop step responses and the singular values of S and T.

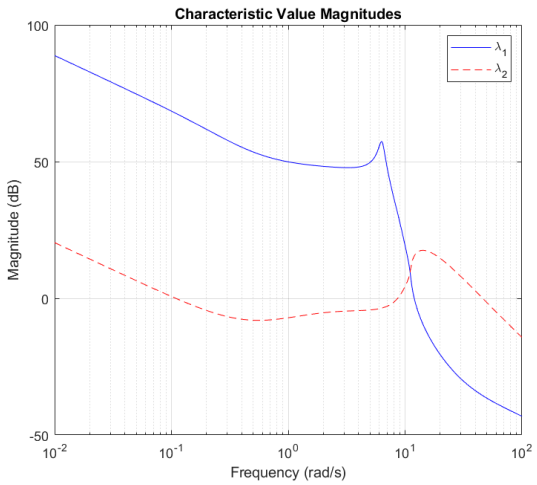
Re-running the code (R2023b)

Both scripts run, and their plots are in `figures/`. However, the closed loop formed in each script, `CL = feedback(K_total, eye(n))`, is **unstable**: its largest pole real part is $+0.34$ for AIRC and $+44.1$ for TGEN. The step-response plots diverge accordingly. Two things look like the cause:

- the characteristic-loci check in the scripts only measures the distance to -1 and does not count encirclements;
- the sign of $K_h = -\operatorname{Re}(UV^*)$ may need to be the opposite.

I left the code as submitted rather than change it after the fact.





Left: characteristic loci of GK_h for AIRC. Right: characteristic-value magnitudes for TGEN.